



EDU TANTR

SECURE DOCUMENT STORAGE SYSTEM

AWS-Based Secure Cloud Document Storage

CLOUD COMPUTING INTERNSHIP PROJECT REPORT

Prepared By

RAKSHITA S K S

Cloud Computing Intern

EDU TANTR

Mentor

KUMAR D

INTERNSHIP OVERVIEW

This project was developed as part of the Cloud Computing Internship at Edutantr. The internship provided practical exposure to cloud services, AWS infrastructure, serverless computing, storage, security, and cloud-based application development.

DETAIL	INFORMATION
Internship Domain	Cloud Computing
Project Title	Secure Document Storage System
Cloud Platform	Amazon Web Services (AWS)
Project Type	Cloud-Based Application
Role	Cloud Computing Intern

TABLE OF CONTENTS

S.NO.	CONTENT	PAGE
1	INTRODUCTION	1
2	PROBLEM STATEMENT	2
3	OBJECTIVES	2
4	AWS SERVICES AND TECHNOLOGIES USED	3
5	SYSTEM ARCHITECTURE	4
6	SYSTEM WORKFLOW	5
7	IMPLEMENTATION	6
8	SECURITY CONFIGURATION	8
9	TESTING AND RESULTS	10
10	OUTPUT	13
11	CHALLENGES AND SOLUTION	16
12	KEY LEARNINGS	17
13	CONCLUSION	18

1.INTRODUCTION

The Secure Document Storage System was developed as part of my Cloud Computing Internship at Edutantr using Amazon Web Services (AWS). The main purpose of the project was to create a secure cloud-based environment for storing documents using Amazon S3.

I configured the S3 storage environment with appropriate access permissions and server-side encryption to protect the stored documents. I also worked with AWS IAM to control access to the resources and used AWS Lambda and CloudWatch as part of the serverless processing and monitoring setup.

Through this project, I gained practical experience in creating and configuring AWS resources, managing permissions, securing stored data, testing the configuration and checking the execution details through AWS monitoring tools.

2.PROBLEM STATEMENT

- Storing important documents without proper access control and encryption can expose them to unauthorized access and security risks. A cloud storage system should provide controlled access to documents while protecting the stored data.
- To address this, I developed a Secure Document Storage System using Amazon S3. The project focuses on secure document storage by configuring IAM permissions and server-side encryption and verifying the configured resources through AWS services.

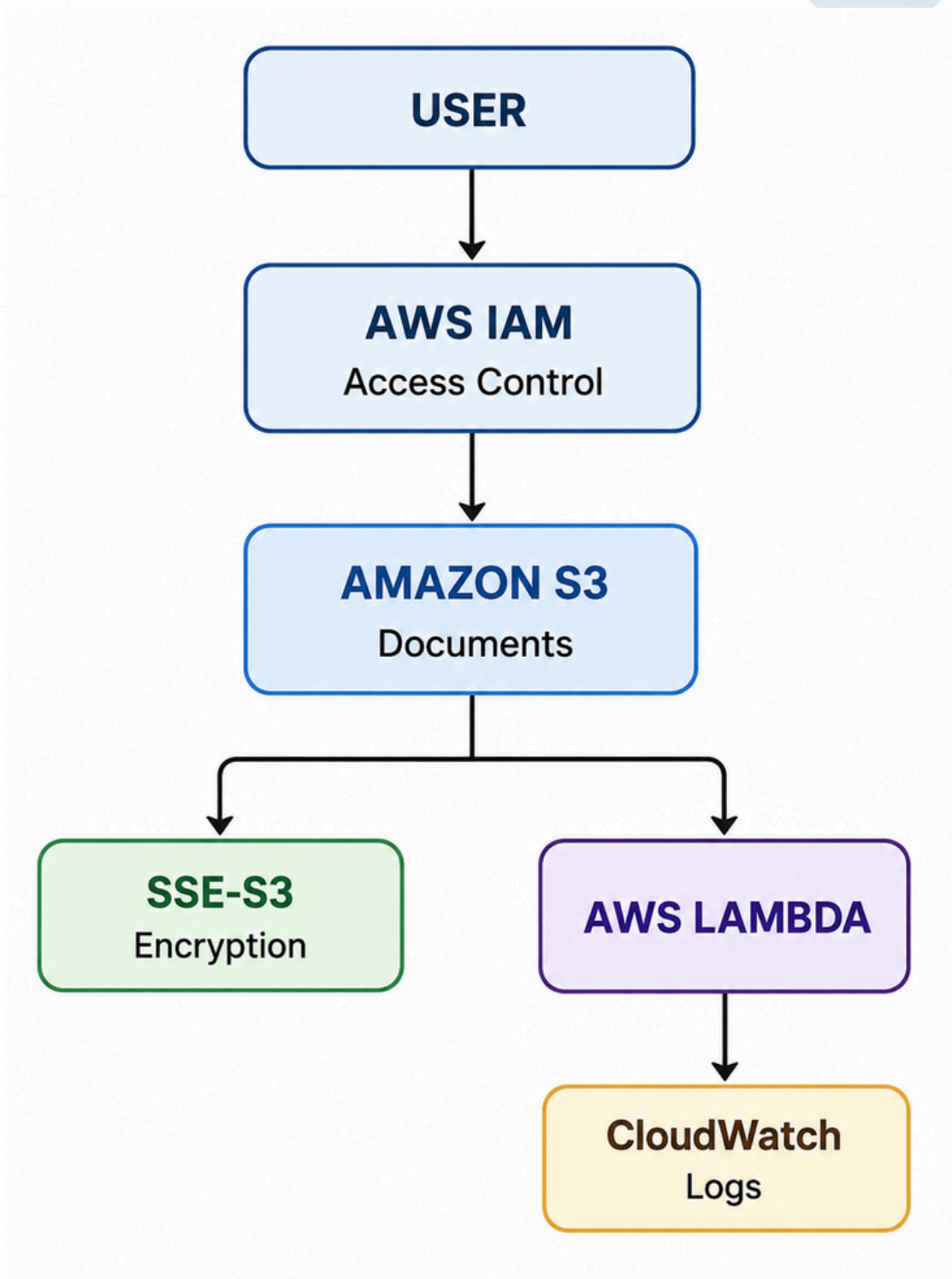
3.OBJECTIVES

- To create a secure document storage system using Amazon S3.
- To upload and store documents in an S3 bucket.
- To configure controlled access using AWS IAM.
- To enable server-side encryption for stored documents.
- To configure the required AWS services for the project.
- To verify the configured resources through testing and monitoring.
- To understand practical cloud storage and security concepts.

4.AWS SERVICES AND TECHNOLOGIES USED

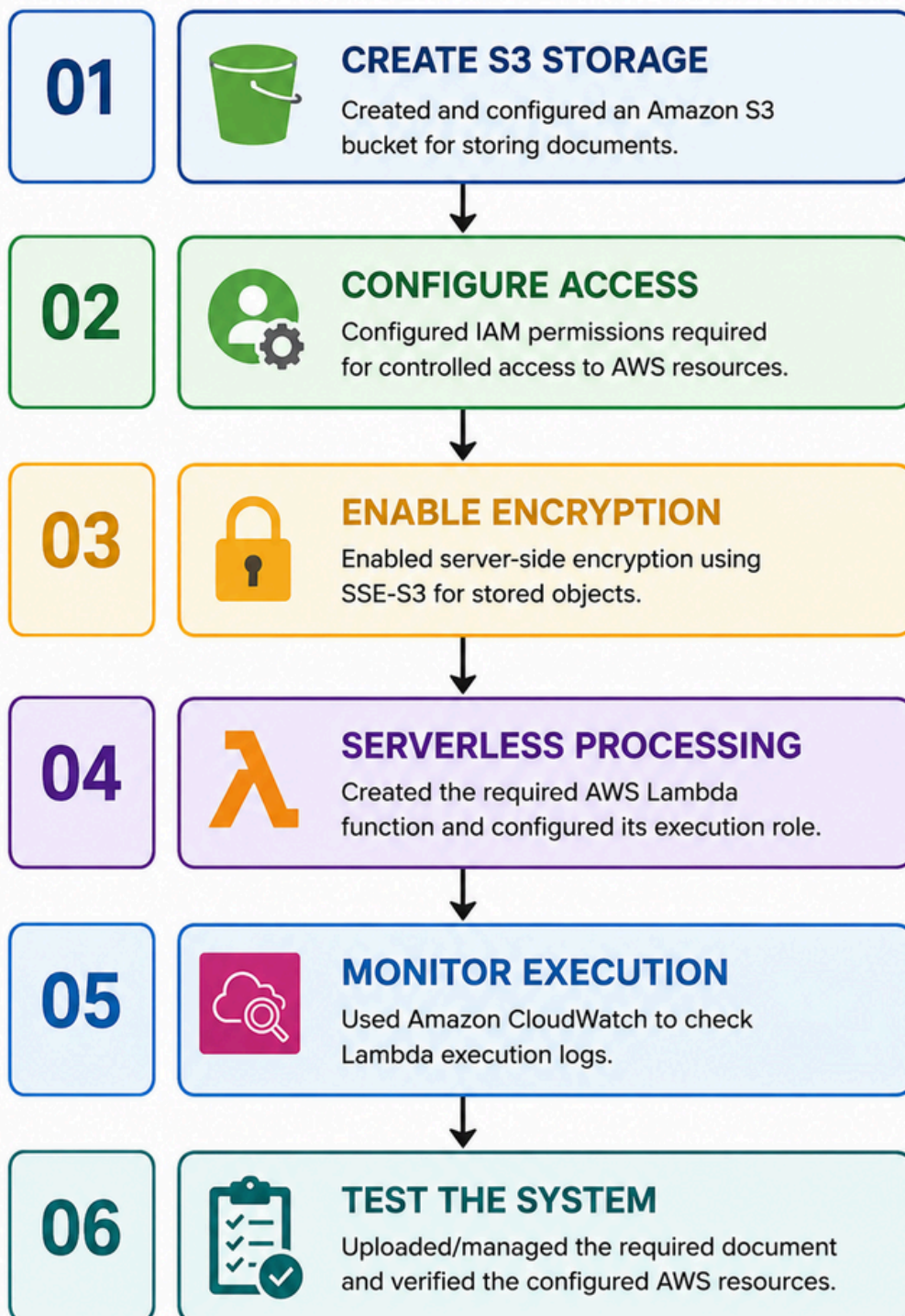
SERVICE/ TECHNOLOGY	PURPOSE	PRACTICAL USE
Amazon S3	Cloud document storage	Created the storage environment and stored documents
AWS IAM	Access control	Created/configured permissions for AWS resources
AWS Lambda	Serverless processing	Used for the project's serverless processing requirement
Amazon CloudWatch	Monitoring	Checked Lambda execution logs
SSE-S3	Data protection	Enabled server-side encryption for stored objects
Python	Programming	Used for the Lambda function implementation

5.SYSTEM ARCHITECTURE



6.SYSTEM WORKFLOW

The Secure Document Storage System was developed using AWS cloud services to store documents securely, control access, protect stored data, and monitor the configured resources.

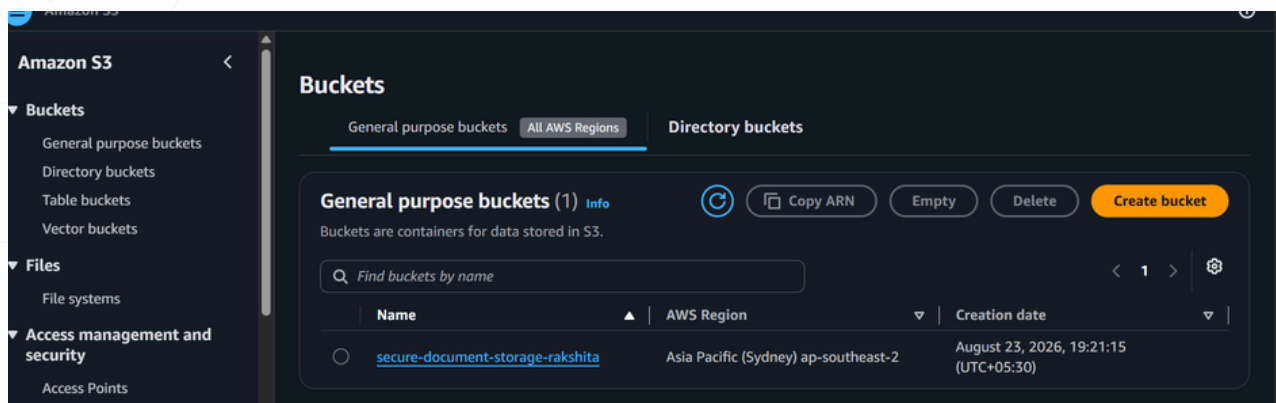


7.IMPLEMENTATION

The Secure Document Storage System was implemented using AWS services to provide secure cloud storage, controlled access, encryption, and serverless processing.

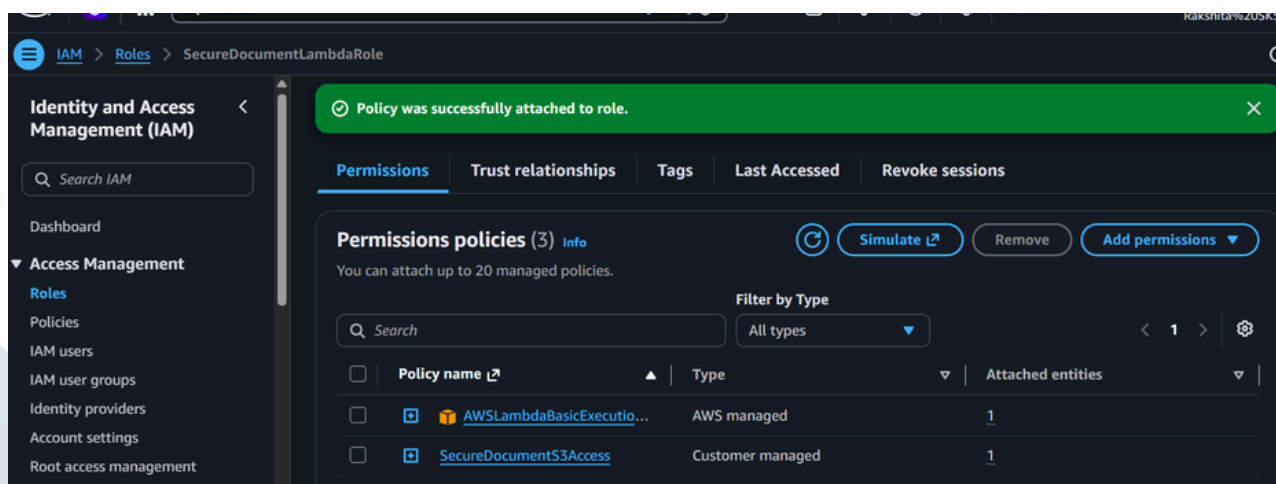
7.1.AMAZON S3 CONFIGURATION

An Amazon S3 bucket was created to store the required documents. The bucket was configured as the main cloud storage location for the system, and the required document was uploaded successfully.



7.2.IAM ACCESS CONTROL

AWS IAM was configured to provide controlled access to the AWS resources used in the project. The required permissions were assigned through the appropriate IAM configuration.



7.3.SERVER-SIDE ENCRYPTION

Server-side encryption using SSE-S3 was enabled for the stored objects. This provides encryption of the document while it is stored in Amazon S3.

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key

When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)

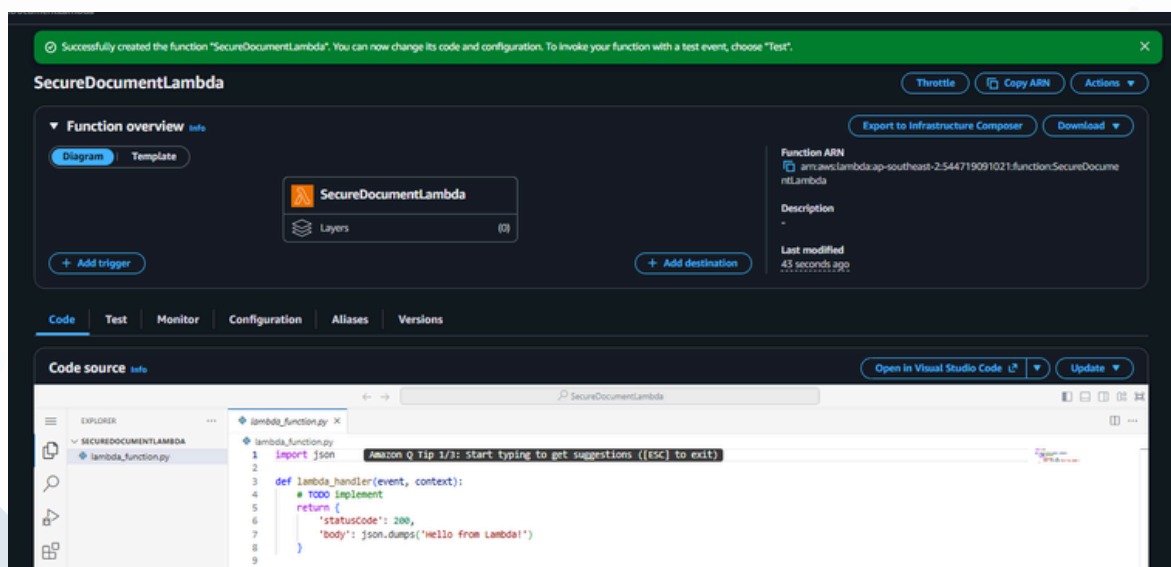
Enabled

Blocked encryption types [Info](#)

Server-Side Encryption with Customer-Provided Keys (SSE-C)

7.4.AWS LAMBDA AND CLOUDWATCH

An AWS Lambda function was configured as part of the serverless workflow. Its execution activity was monitored using Amazon CloudWatch logs to verify the operation of the configured resources.

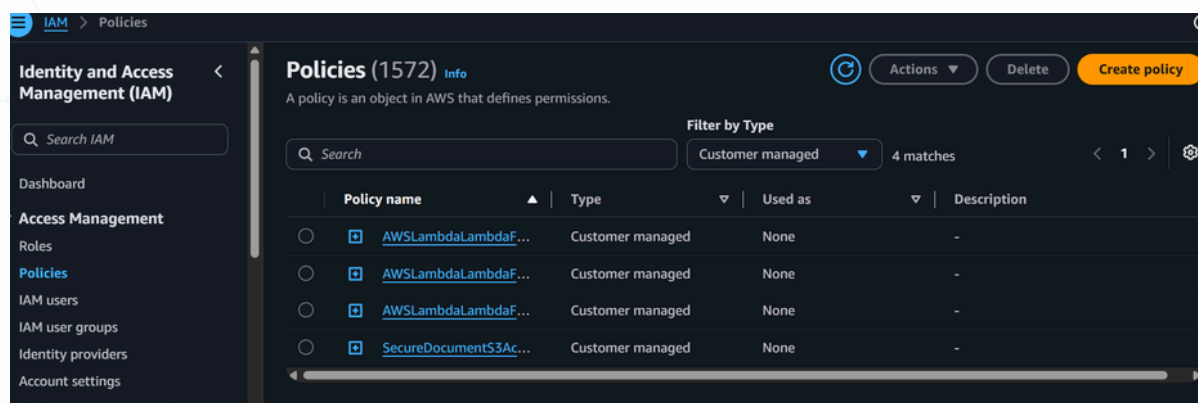


8. SECURITY CONFIGURATION

Security configurations were implemented to protect the documents stored in Amazon S3 and to control access to the AWS resources used in the project. IAM permissions and server-side encryption were configured and verified through the AWS Management Console.

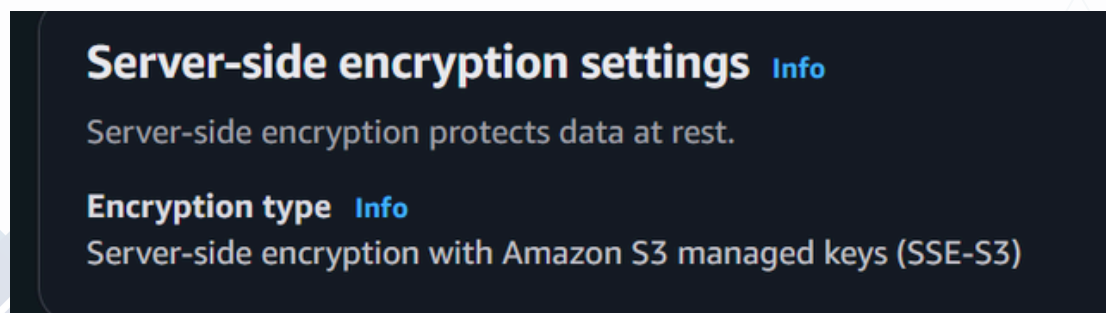
8.1. IAM ACCESS CONTROL

AWS IAM was used to control access to the resources required for the project. An IAM role with the required permissions was configured so that the AWS Lambda function could access the necessary S3 resources securely.



8.2. S3 SERVER-SIDE ENCRYPTION

Server-side encryption using SSE-S3 was enabled for the S3 objects. This protects the stored documents by encrypting the data while it is stored in Amazon S3.



8.3.SECURITY VERIFICATION

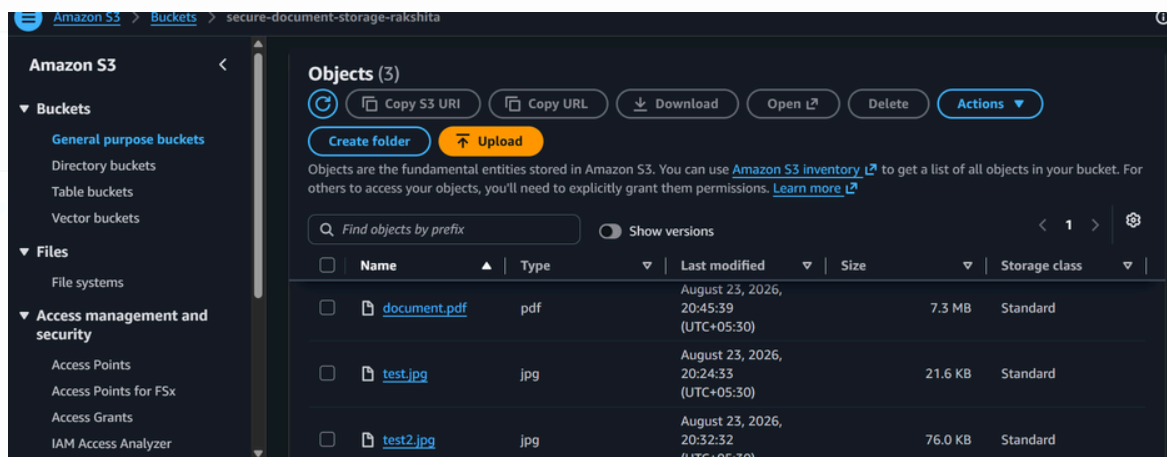
The configured IAM permissions and S3 encryption settings were checked through the AWS Management Console. This verified that the stored documents were protected and that the required AWS resources had controlled access.

9. TESTING AND RESULTS

After completing the AWS configuration, the Secure Document Storage System was tested to verify the storage, access control, encryption and serverless components. The configured resources were checked through the AWS Management Console and the execution activity was monitored using CloudWatch.

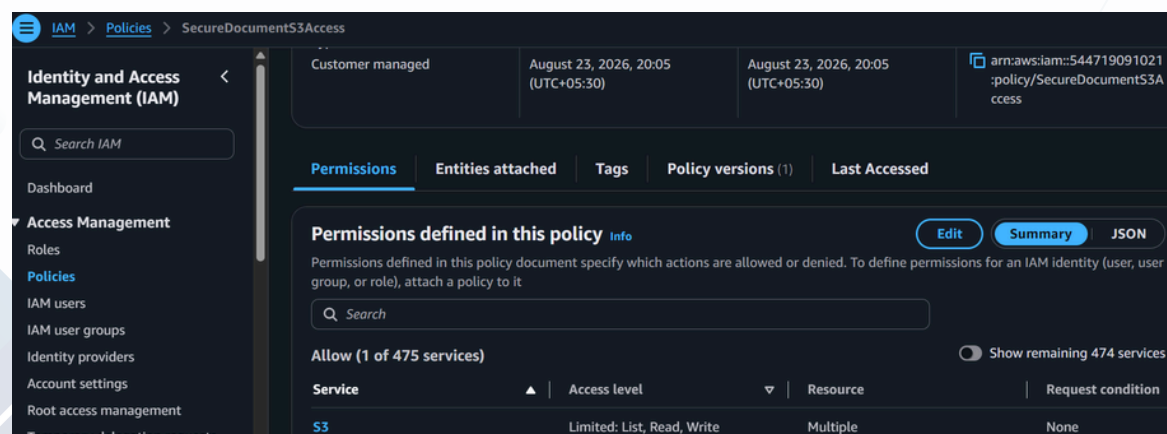
9.1.S3 STORAGE VERIFICATION

The S3 bucket was checked to confirm that the required document was successfully uploaded and available in the configured storage location.



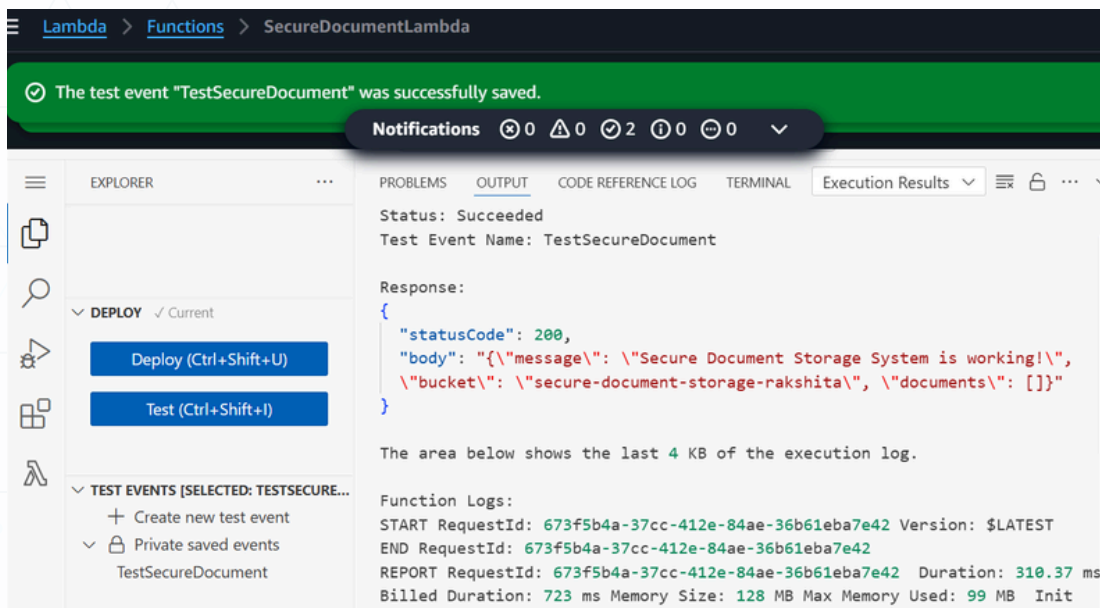
9.2.ACCESS AND SECURITY VERIFICATION

The configured IAM permissions and S3 server-side encryption settings were checked to verify that the required access control and data protection were applied.



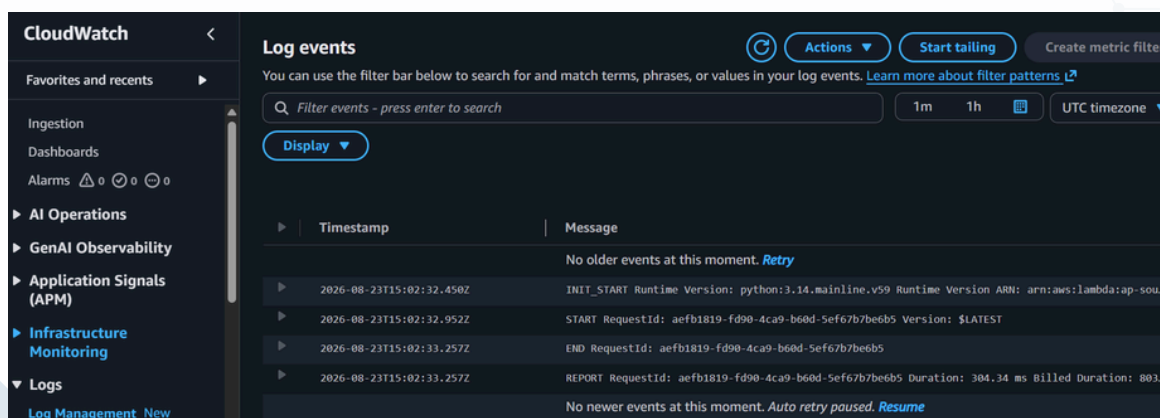
9.3.LAMBDA EXECUTION VERIFICATION

The AWS Lambda function was tested to verify that the configured serverless workflow executed successfully. The execution details were checked through the AWS console.



9.4.CLOUDWATCH MONITORING

The Lambda execution logs were checked in Amazon CloudWatch to verify the execution activity and identify any errors during testing.



RESULT

The testing confirmed that the configured AWS resources were functioning correctly. The document was successfully stored in Amazon S3, the required access and encryption settings were configured, and the Lambda execution could be monitored through CloudWatch.

10.OUTPUT

The final output of the Secure Document Storage System is a secure cloud-based storage environment in Amazon S3. The configured IAM access control and SSE-S3 encryption protect the stored document, while the AWS services used in the project were successfully configured and verified.

10.1.S3 OUTPUT

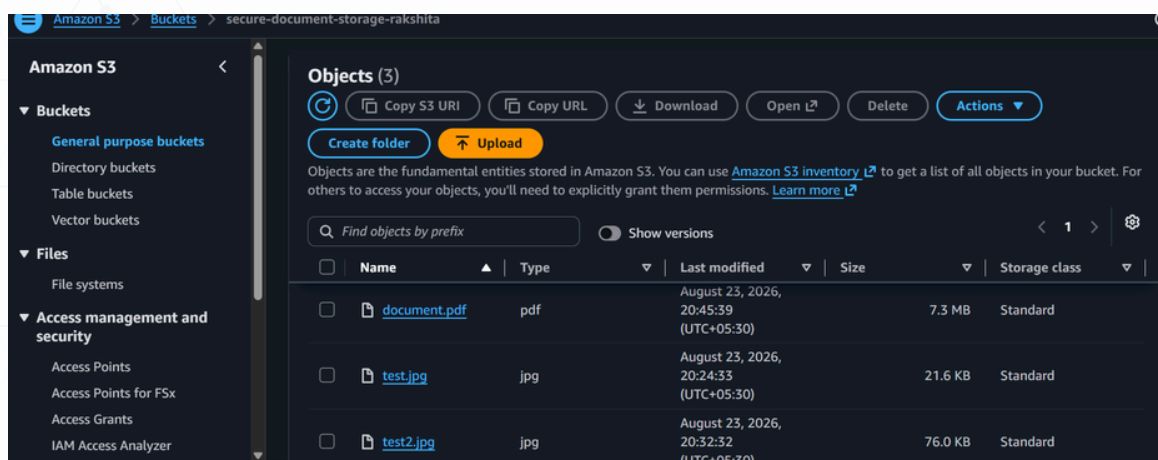


Figure 10.1: Document successfully stored in Amazon S3

10.2.ENCRYPTION

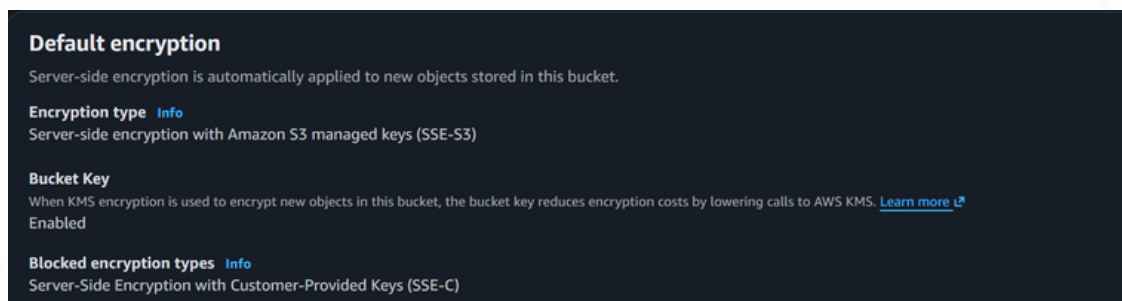


Figure 10.2: SSE-S3 server-side encryption enabled for the stored document

10.3.LAMBDA+S3 TRIGGER

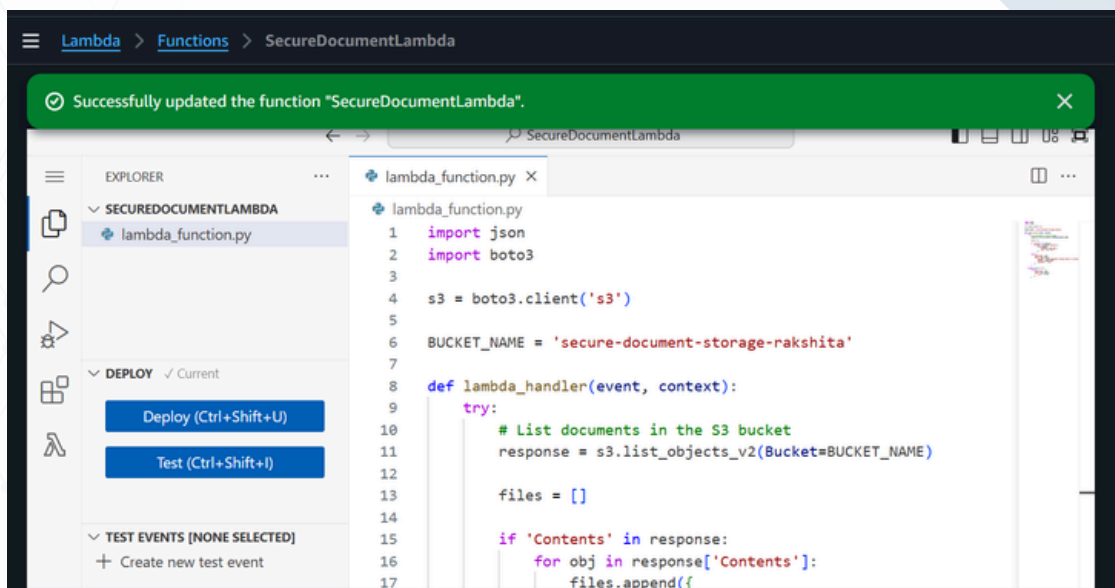
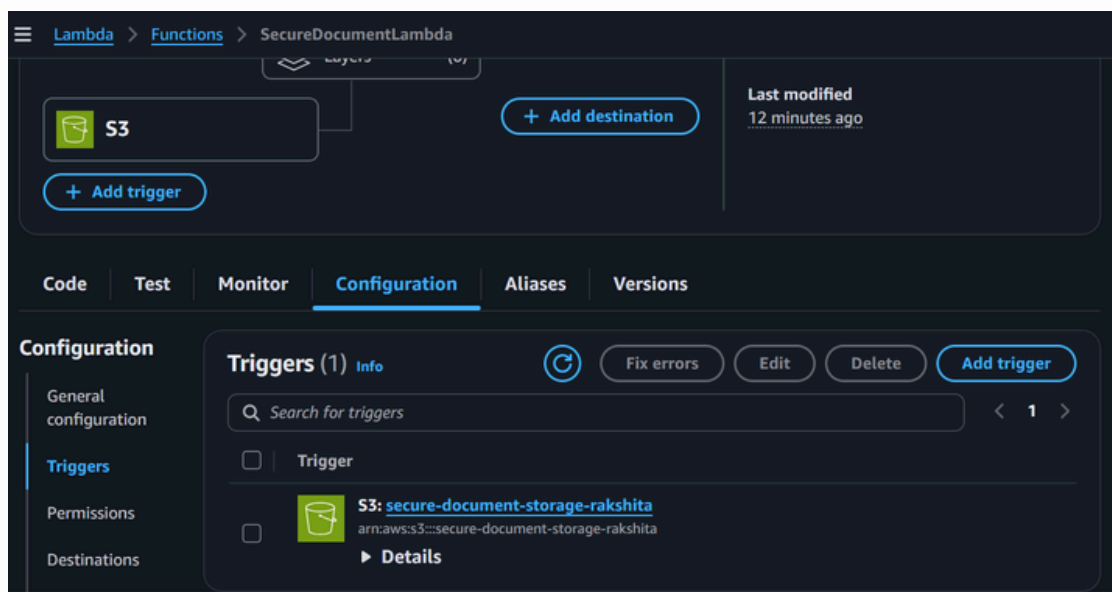


Figure 10.3:S3 trigger configured to invoke the AWS Lambda function.



10.4.EXECUTION OF LAMBDA FUNCTION

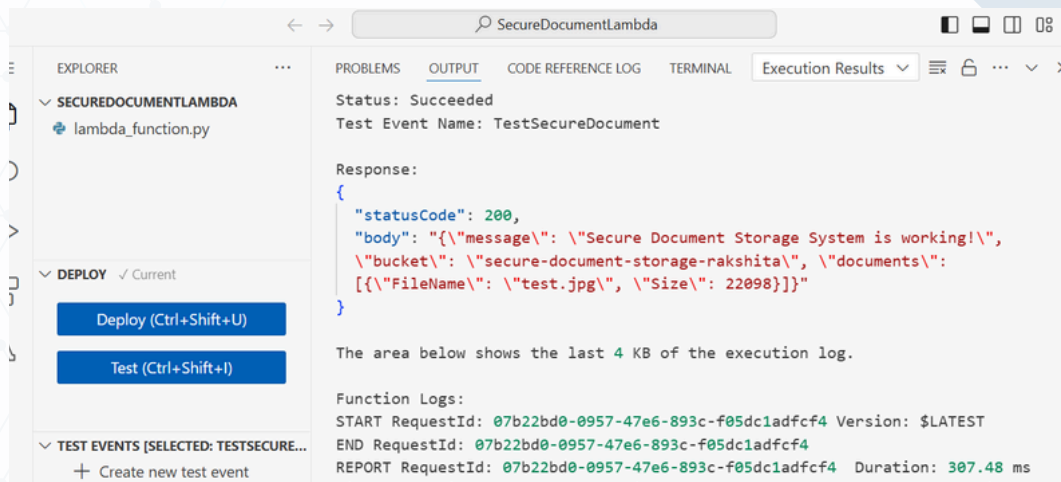


Figure 10.4: Successful execution of the AWS Lambda function

10.5.CLOUDWATCH LOGS

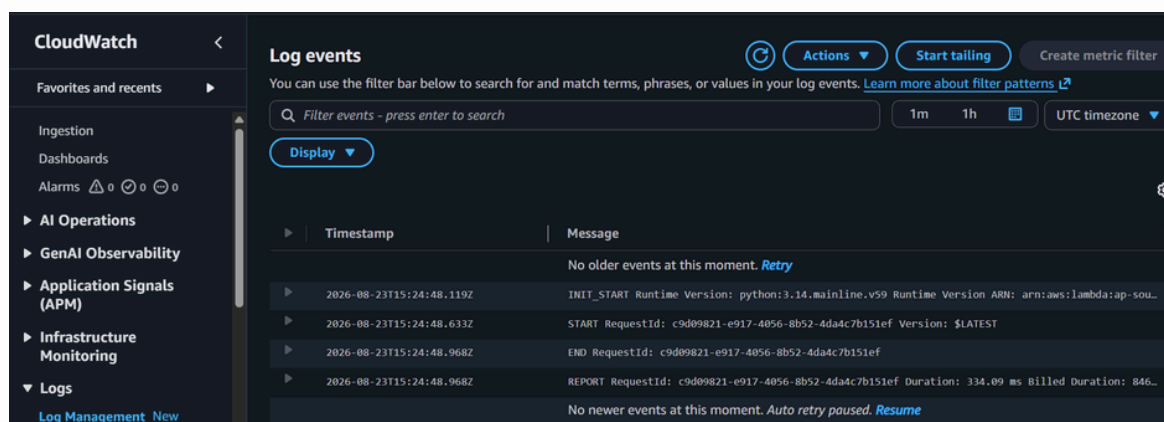


Figure 10.5: AWS CloudWatch logs showing Lambda execution activity



FINAL OUTPUT:

The document was successfully stored in Amazon S3 with the configured access control and server-side encryption.

11. CHALLENGES AND SOLUTION

CHALLENGE FACED	SOLUTION
Configuring the S3 trigger with Lambda	Configured the correct S3 bucket trigger and tested the Lambda function.
Setting the required IAM permissions	Created and configured the required IAM execution role and permissions.
Packaging required Python dependencies	Packaged the required Pillow dependency with the Lambda deployment package.
S3 object key mismatch during testing	Checked the actual S3 object key and corrected the Lambda test event.
Verifying Lambda execution	Used CloudWatch logs to monitor and verify Lambda execution.

12.KEY LEARNINGS

Through this project, I gained practical experience in configuring and using AWS services for secure cloud storage. I learned how different AWS services can be integrated to build and manage a cloud-based application.

- Learned to create and manage Amazon S3 buckets and objects.
- Learned to configure AWS IAM roles and permissions for controlled access.
- Learned to enable SSE-S3 server-side encryption for stored data.
- Gained practical experience in creating and configuring AWS Lambda functions.
- Learned to connect Amazon S3 with AWS Lambda using triggers.
- Learned to monitor Lambda execution using Amazon CloudWatch.
- Improved my ability to test, troubleshoot and configure AWS resources.
- Understood the importance of secure access and proper resource management in cloud applications.

12.1.REFERENCES

1. Amazon Web Services — Amazon S3 Documentation
2. Amazon Web Services — AWS IAM Documentation
3. Amazon Web Services — AWS Lambda Documentation
4. Amazon Web Services — Amazon CloudWatch Documentation

13.CONCLUSION

The Secure Document Storage System provided me with practical experience in implementing a cloud-based solution using AWS. I created and configured Amazon S3 for document storage, managed access using IAM, enabled SSE-S3 encryption, and configured AWS Lambda with an S3 trigger.

I also used Amazon CloudWatch to monitor Lambda execution and performed testing to verify the configured resources. The project helped me understand how AWS services can work together to provide secure and reliable cloud storage.

Overall, this project improved my practical knowledge of AWS S3, IAM, Lambda, CloudWatch, encryption, access control and cloud resource management.